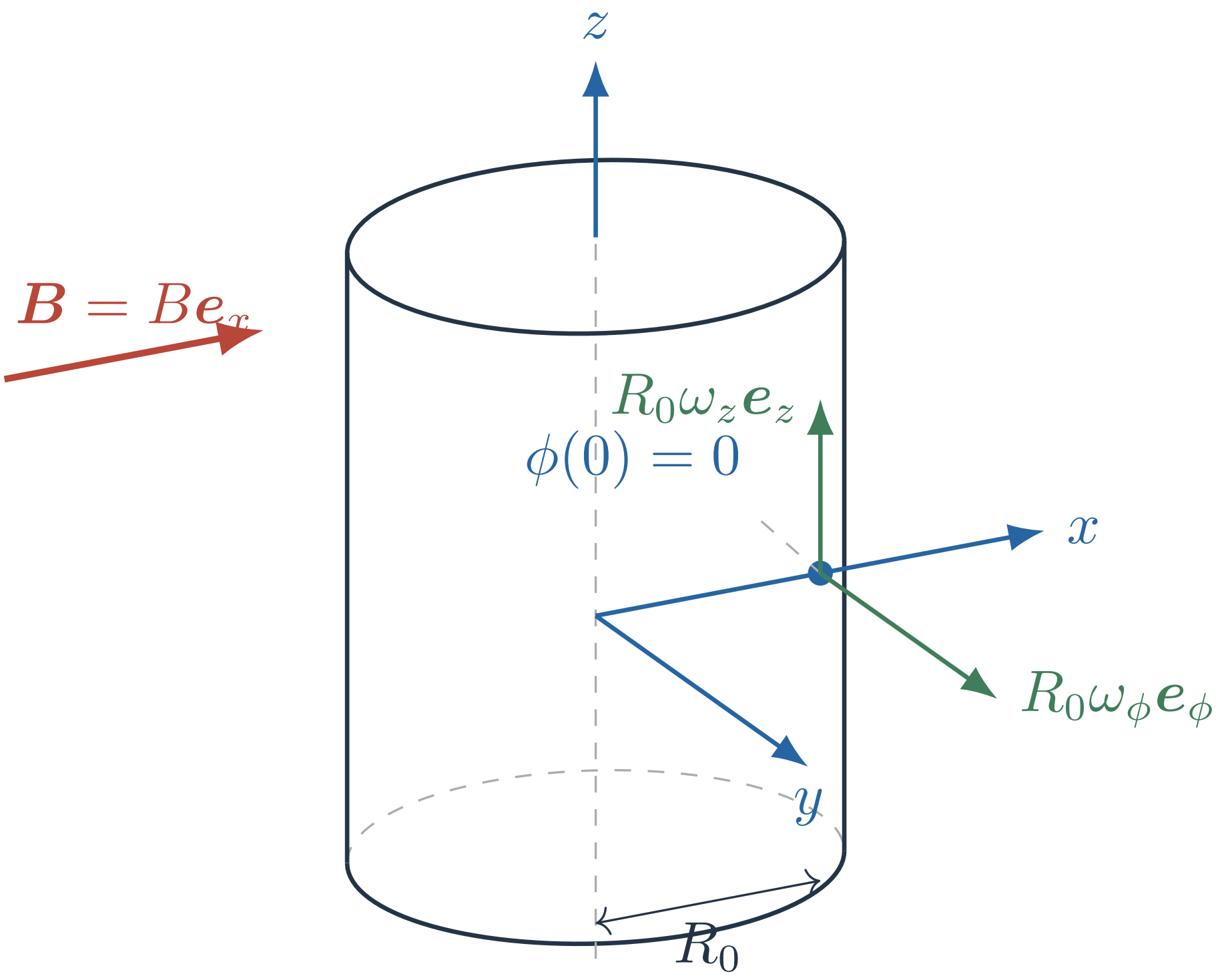
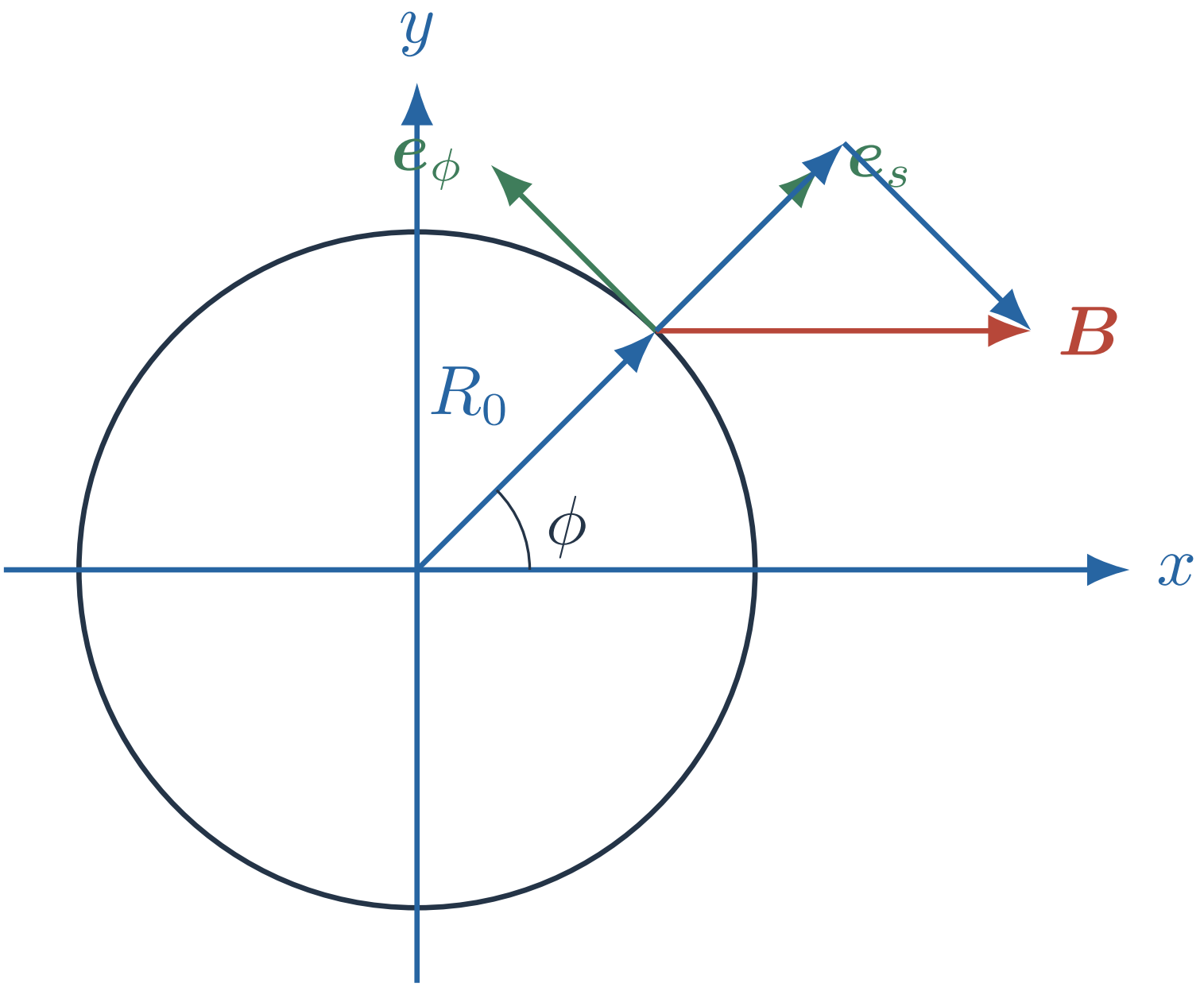
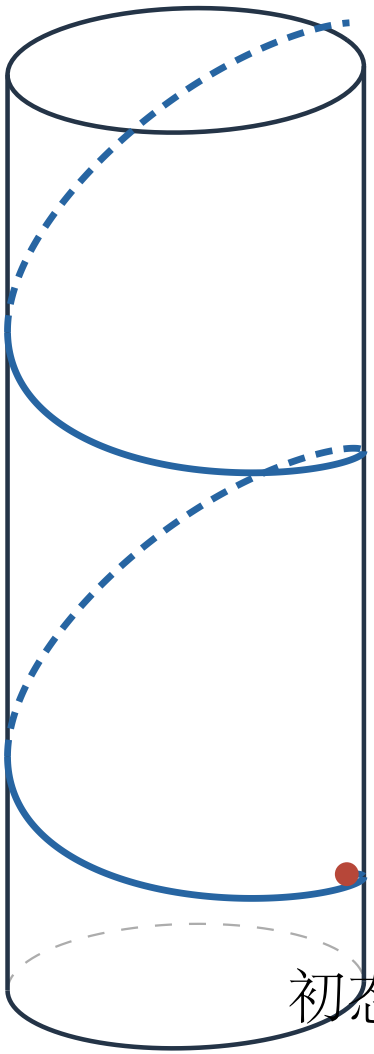


光滑圆柱面，径向约束维持半径



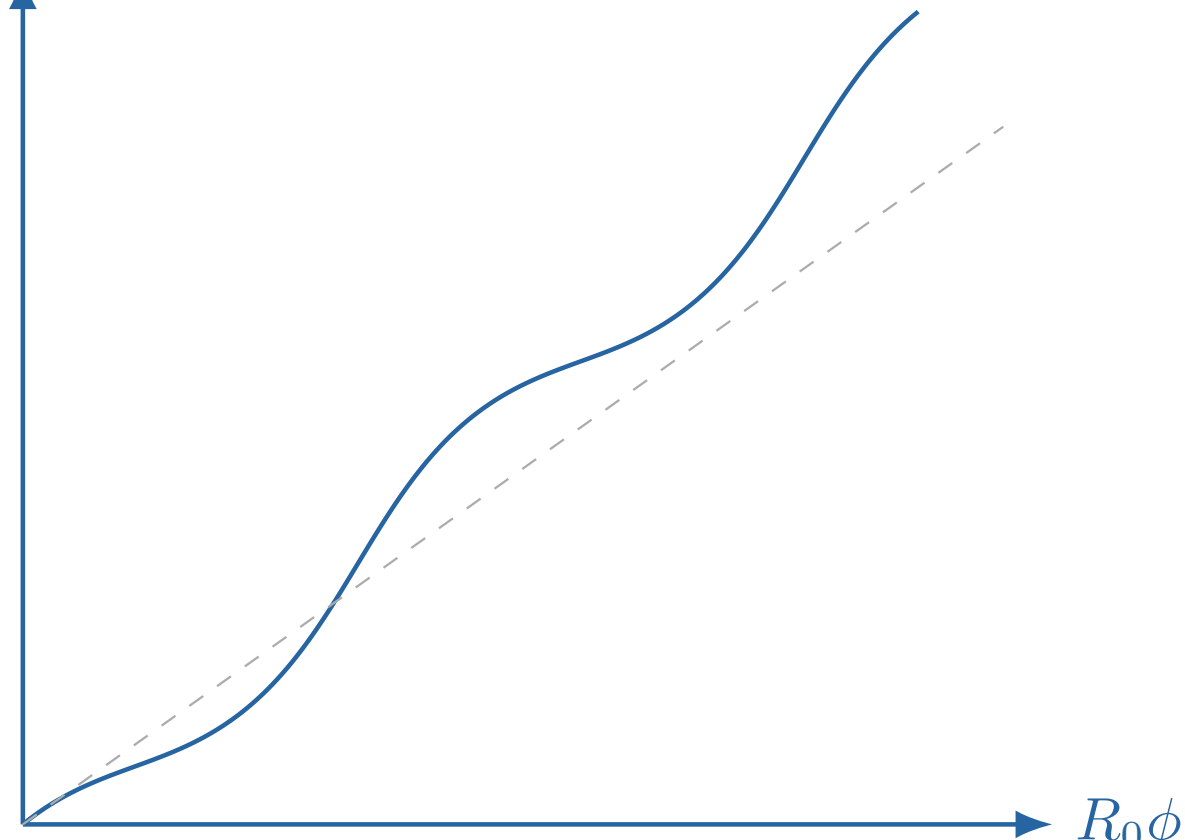
$$\boldsymbol{B} = B\cos\phi\,\boldsymbol{e}_s - B\sin\phi\,\boldsymbol{e}_\phi$$

三维绕轴轨迹

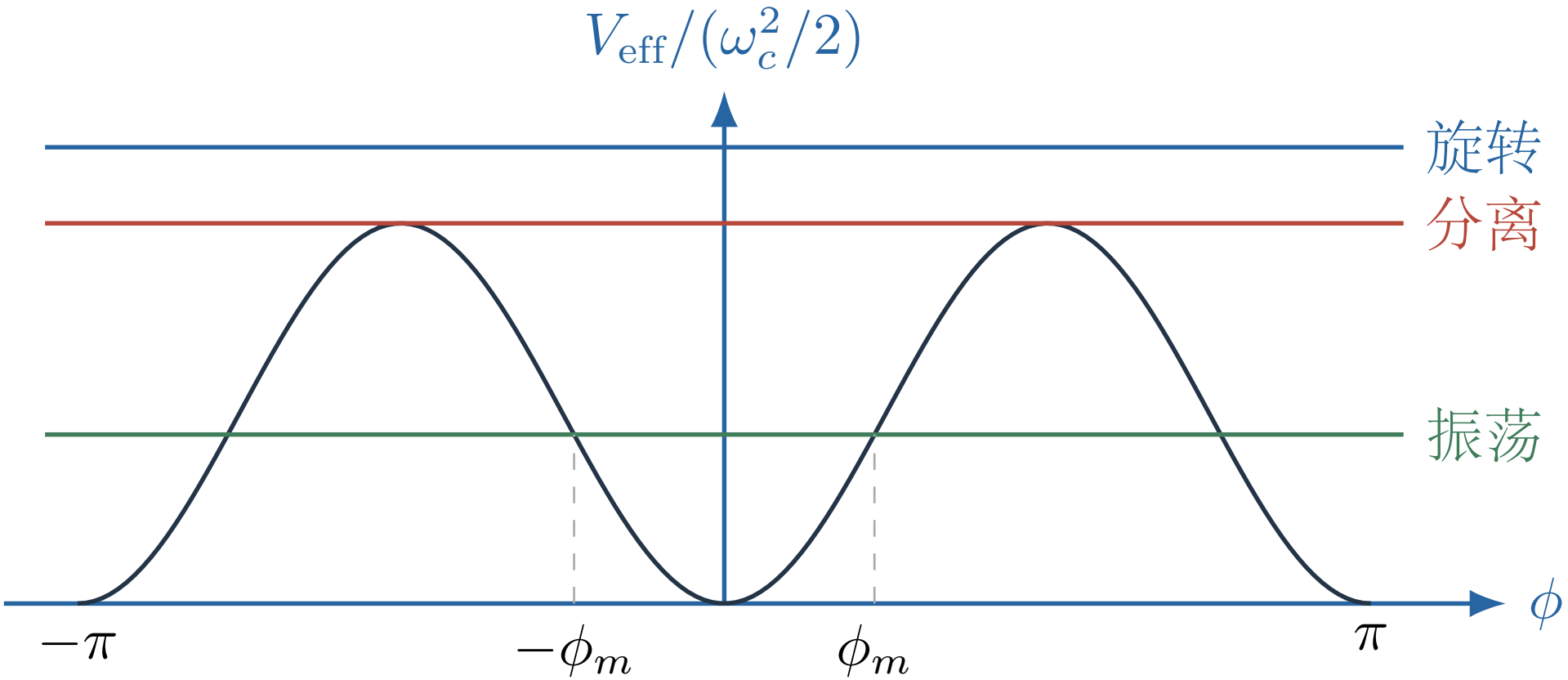


同一轨迹展开

虚线仅为平均斜率参照



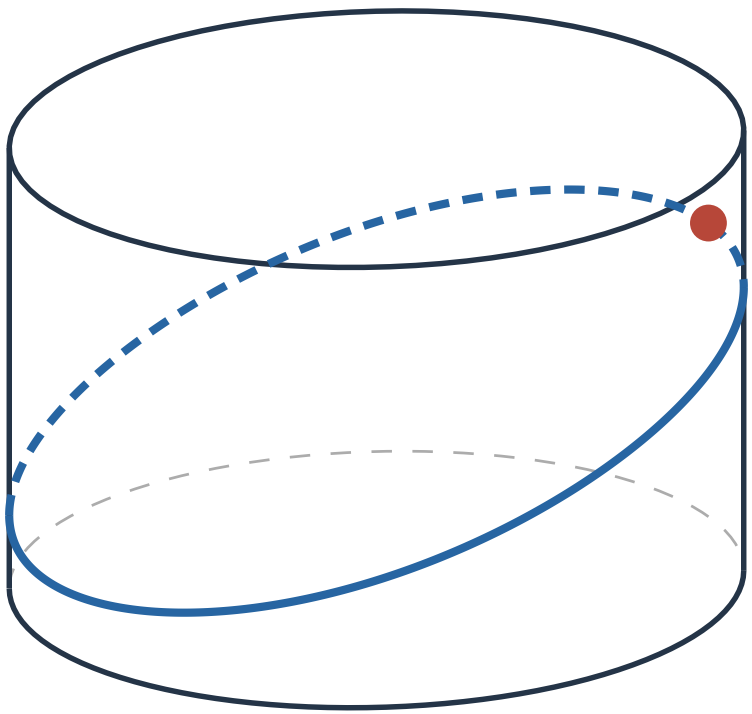
示例 $\omega_\phi = 1,\ \omega_z = 0.6,\ \omega_c = 0.3$；长度以 R_0 为单位



$$\omega_z = 0\colon\ V_{\text{eff}} = \tfrac{1}{2}\omega_c^2 \sin^2\phi$$

弱场：完整绕轴

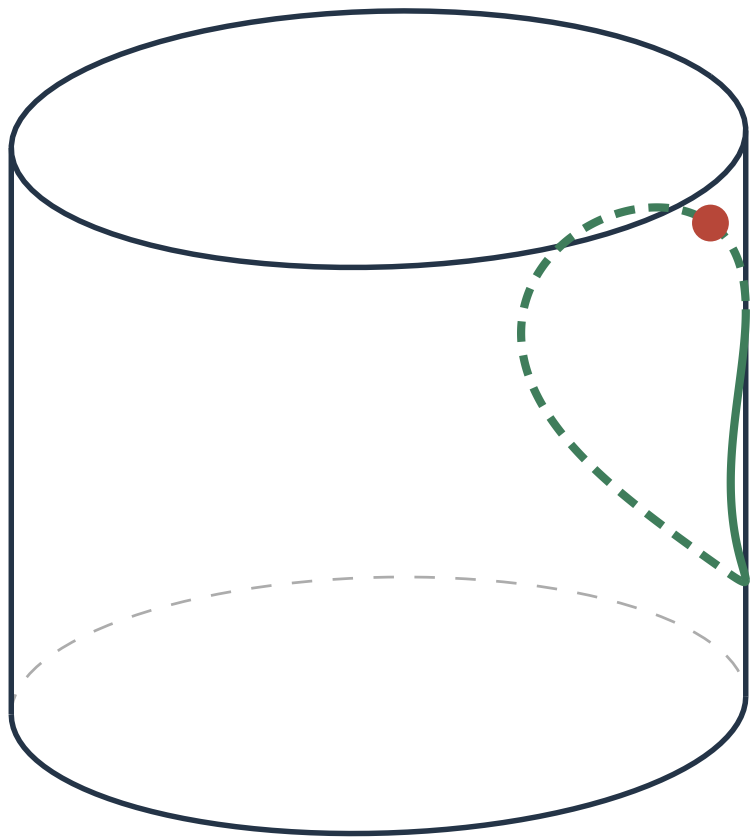
$|\omega_c/\omega_\phi| = 0.5$



一周后回到初始高度

强场：有限角区往返

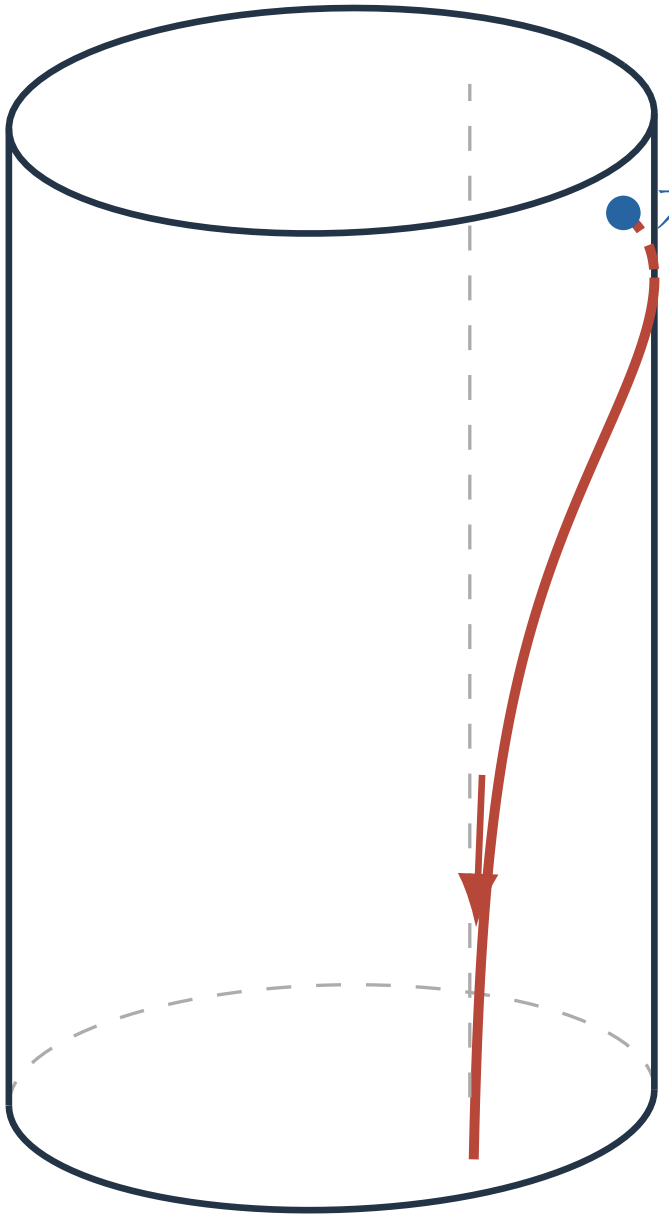
$|\omega_c/\omega_\phi| = 1.5$



$\phi_m = \arcsin(2/3)$

实线为前侧，虚线为背侧轨迹

临界示例 $\omega_c = \omega_\phi = \omega > 0,\ \omega_z = 0$



$$\sin\phi = \tanh\omega t$$

$$z - z_0 = -R_0 \ln\cosh\omega t$$

$\phi \to \pi/2$

$z \to -\infty$，不闭合